

## **DSO 579 Mini-Project Report**

### **Comparative Analysis of Competitiveness in European Football Leagues and MLS**

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## **1. Introduction**

### **Objective:**

This project aims to compare the competitiveness between European football leagues and Major League Soccer (MLS) using team-level performance data, focusing on fixture-based and squad performance metrics.

### **Research Question:**

What key team-level performance metrics distinguish the competitive balance of European football leagues from that of Major League Soccer (MLS)?

### **Assumption:**

A foundational assumption of this analysis is that European leagues are inherently more competitive than MLS—a view that is widely accepted based on historical performance, structural differences, and population consensus. European teams are, therefore, labeled as “competitive” (1) and MLS teams as “non-competitive” (0), allowing us to use supervised learning techniques to identify the performance metrics that drive this difference.

### **Background and Rationale:**

Competitive balance is widely regarded as a cornerstone of an engaging and sustainable sports league. When teams have a relatively equal chance of winning, fan interest and matchday excitement tend to increase, driving higher viewership, attendance, and financial performance. As noted by András and Havran (2018), leagues with tighter point distributions and more frequent shifts in team rankings tend to maintain greater fan engagement over time. In contrast, persistent dominance by a few clubs can lead to predictability and declining interest. European football leagues are often perceived as more competitive than Major League Soccer (MLS), not necessarily due to parity, but because of their open league structures, promotion and relegation systems, and access to international competitions like the UEFA Champions League. These factors raise the performance stakes and incentivize broader competition across all tiers (UEFA, 2022; Deloitte, 2023). Meanwhile, MLS operates under a closed-league model with centralized control and no relegation, which, while financially stable, is sometimes criticized for limiting competitive pressure. This perceived difference underpins our assumption that European leagues are more competitive and serves as the foundation for our supervised learning framework.

## 2. Methodology

### Data Sources and Preparation:

#### • Datasets Used:

I combined team-level data from fixtures, squad performance statistics, and league standings for European football leagues and MLS across multiple seasons. After standardizing key columns (League, Season, Team) and cleaning the data, I derived several performance metrics (e.g., average goals scored/conceded, win percentages, close match percentages, goals per 90, assists per 90, etc.). All the data was scraped from Fbref.

#### • Assumption and Binary Outcome:

Based on a widely accepted notion, I assume that European leagues are more competitive than MLS. Hence, I define a binary variable competitive:

- European teams are labeled as **1 (competitive)**
- MLS teams are labeled as **0 (non-competitive)**

#### • Qualified Teams:

In addition to the full team-level dataset, I extracted a subset of “qualified” teams using performance indicators found in the *Notes* column of the standings data (e.g., qualification for European competitions in Europe and playoff-related notes for MLS). This subset is intended to capture higher-quality teams and focus the analysis on cases where performance differences are most pronounced.

### Modeling Approaches:

I built two sets of models for both supervised methods—logistic regression and decision trees—to analyze the competitive differences:

#### I. Using All Teams:

##### • Logistic Regression:

I applied logistic regression (with feature selection via RFECV and cross-validation) on the full team-level dataset. This model predicts competitive status (1 for European, 0 for MLS) based on all available performance metrics, providing coefficient estimates that indicate the impact of each feature in its raw units.

##### • Decision Tree:

I built a decision tree using the greedy method with the Gini index to split the data. Cost complexity pruning (using cross-validation for optimal `ccp_alpha`) was then applied to reduce overfitting while maintaining interpretability. The resulting tree yields clear decision rules that reflect performance thresholds differentiating European teams from MLS teams.

## **II. Using Qualified (High-Quality) Teams:**

### **• Logistic Regression:**

For this analysis, I filtered the dataset to only include teams that meet a predefined “qualified” criterion (i.e., teams that have reached continental competitions or playoffs). Logistic regression was then applied to this subset, and I again used RFECV with cross-validation to select the most predictive features. The resulting coefficient estimates and associated statistics provide insights into which performance metrics are most indicative of competitive teams among the higher-quality subset.

### **• Decision Tree:**

Similarly, a decision tree was constructed and pruned for the qualified teams. The tree’s decision rules—based on features like win percentages, goals conceded, and squad statistics—offer a more focused view of what differentiates top European teams from MLS teams.

### **Cross-Validation and Evaluation Metrics:**

For both the logistic regression and decision tree models, k-fold (5-fold) cross-validation was used to ensure that the models generalize. I evaluated models based on accuracy and, for logistic regression, interpreted feature coefficients (or odds ratios) and statistical significance. The decision tree model was evaluated both by its cross-validated accuracy and by visually inspecting the tree structure.

## **3. Results**

### **Full-Team Analysis:**

#### **• Logistic Regression Results:**

- The model, based on all teams, achieved an average cross-validated accuracy of approximately 88–90%.
- Key predictors included metrics such as away win percentage, close match percentage at home/away, and squad-level indicators like average age.

- The detailed stats model summary provided below shows significant coefficients (with p-values < 0.05) for several features, confirming that these metrics are closely associated with being competitive (European).

| Logit Regression Results  |                  |                   |            |       |           |          |
|---------------------------|------------------|-------------------|------------|-------|-----------|----------|
| =====                     |                  |                   |            |       |           |          |
| Dep. Variable:            | competitive      | No. Observations: | 1510       |       |           |          |
| Model:                    | Logit            | Df Residuals:     | 1485       |       |           |          |
| Method:                   | MLE              | Df Model:         | 24         |       |           |          |
| Date:                     | Tue, 08 Apr 2025 | Pseudo R-squ.:    | 0.4714     |       |           |          |
| Time:                     | 09:01:15         | Log-Likelihood:   | -328.58    |       |           |          |
| converged:                | False            | LL-Null:          | -621.57    |       |           |          |
| Covariance Type:          | nonrobust        | LLR p-value:      | 2.025e-108 |       |           |          |
| =====                     |                  |                   |            |       |           |          |
|                           | coef             | std err           | z          | P> z  | [0.025    | 0.975]   |
| -----                     |                  |                   |            |       |           |          |
| const                     | 15.6014          | 2.861             | 5.454      | 0.000 | 9.995     | 21.208   |
| avg_scored_home           | -2.0210          | 1.566             | -1.291     | 0.197 | -5.090    | 1.048    |
| avg_conceded_home         | 15.3312          | 12.550            | 1.222      | 0.222 | -9.266    | 39.929   |
| home_win_pct              | 0.0140           | 0.015             | 0.919      | 0.358 | -0.016    | 0.044    |
| home_close_match_pct      | -0.0342          | 0.010             | -3.545     | 0.000 | -0.053    | -0.015   |
| avg_scored_away           | -0.5859          | 1.577             | -0.371     | 0.710 | -3.678    | 2.506    |
| avg_conceded_away         | 13.7863          | 12.524            | 1.101      | 0.271 | -10.760   | 38.333   |
| away_win_pct              | 0.0591           | 0.015             | 3.986      | 0.000 | 0.030     | 0.088    |
| away_close_match_pct      | -0.0403          | 0.009             | -4.328     | 0.000 | -0.059    | -0.022   |
| overall_avg_conceded      | -28.2369         | 25.065            | -1.127     | 0.260 | -77.364   | 20.890   |
| players_used              | -0.0374          | 0.035             | -1.081     | 0.280 | -0.105    | 0.030    |
| avg_age                   | -0.4556          | 0.091             | -4.985     | 0.000 | -0.635    | -0.276   |
| goals                     | 0.6611           | 6.85e+04          | 9.65e-06   | 1.000 | -1.34e+05 | 1.34e+05 |
| goals_per90               | -40.5351         | 24.667            | -1.643     | 0.100 | -88.882   | 7.811    |
| assists                   | -0.8279          | nan               | nan        | nan   | nan       | nan      |
| assists_per90             | 13.9978          | 29.993            | 0.467      | 0.641 | -44.787   | 72.783   |
| goals_and_assists         | -0.1658          | nan               | nan        | nan   | nan       | nan      |
| goals_and_assists_per90   | 27.5028          | 24.533            | 1.121      | 0.262 | -20.581   | 75.587   |
| goals_excluding_pk        | 0.8071           | 5.87e+04          | 1.38e-05   | 1.000 | -1.15e+05 | 1.15e+05 |
| goals_excluding_pk_per90  | -20.7161         | 29.550            | -0.701     | 0.483 | -78.634   | 37.202   |
| penalty_kicks             | -0.1445          | 5.87e+04          | -2.46e-06  | 1.000 | -1.15e+05 | 1.15e+05 |
| penalty_attempts          | 0.0352           | 0.097             | 0.363      | 0.717 | -0.155    | 0.225    |
| yellow_cards              | 0.0263           | 0.008             | 3.428      | 0.001 | 0.011     | 0.041    |
| red_cards                 | -0.0817          | 0.047             | -1.740     | 0.082 | -0.174    | 0.010    |
| goals_and_assists_excl_pk | -9.0880          | 26.382            | -0.344     | 0.730 | -60.796   | 42.619   |
| =====                     |                  |                   |            |       |           |          |

#### • Decision Tree Results:

- The decision tree, grown with the Gini criterion and pruned via cost complexity pruning (optimal ccp\_alpha  $\approx$  0.00314), revealed clear splits along key features such as yellow cards, win percentages, and average age.
- The tree's structure allowed us to see specific thresholds that demarcate European teams from MLS teams.

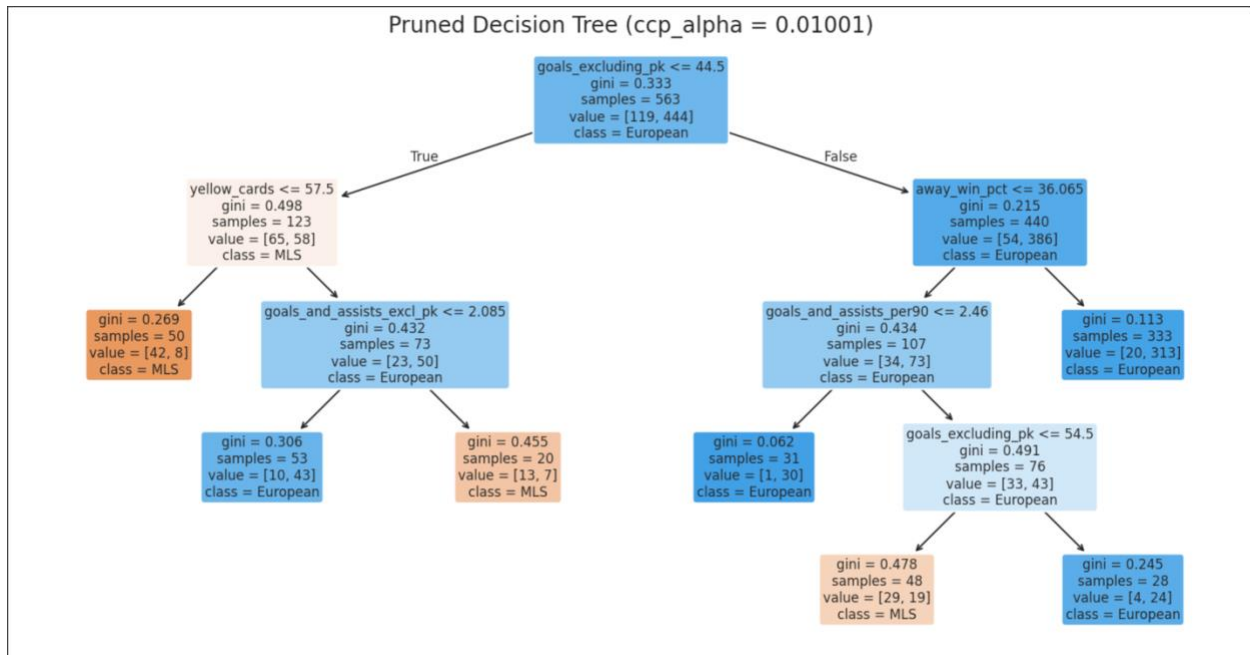
#### Qualified-Team Analysis:

#### • Logistic Regression Results:

- When focusing on the subset of higher-quality (qualified) teams, logistic regression yielded a slightly different set of key features and coefficients, suggesting that the factors driving competitive performance among top-tier teams are nuanced.
- The model maintained strong predictive performance (accuracy around 88–90%), with critical predictors including fine-grained performance metrics like close match percentages and differential scoring measures.

#### • Decision Tree Results:

- The pruned decision tree for qualified teams produced a simpler structure with decision rules that are easier to interpret.
- The rules indicated that certain performance levels (e.g., a minimum away win percentage or a specific threshold for goals without penalty) are particularly discriminative among the highest-quality teams.



#### Interpretation:

Both sets of models support the initial assumption that European leagues display higher competitive balance compared to MLS—yet they also provide insight into *how* this difference manifests. The full-team models capture the broader trends, while the qualified-team models highlight the detailed performance nuances that characterize high-quality teams. For example, European teams tend to have higher win percentages, lower close match percentages, and certain squad characteristics (like optimal average age) that correlate with overall competitive performance.

## 4. Discussion

While the analysis is grounded in the assumption that European leagues are more competitive, the findings shed light on **in what way** this competitiveness is reflected in the data:

- **Performance Metrics:**

Metrics such as win percentages, goals conceded, and scoring efficiency are significant predictors.

- **Threshold Effects:**

The decision tree models reveal discrete performance thresholds that help classify teams into competitive versus non-competitive categories.

- **Qualified Teams vs. All Teams:**

The differences between the full-team and the qualified-team models underscore that even within high-quality teams, there are subtle performance patterns that distinguish European teams from MLS teams.

### **Recommendations:**

Based on these results, potential recommendations for MLS might include improving defensive consistency and optimizing squad composition to better match the performance patterns seen in European leagues.

### **Limitations and Future Work:**

- The analysis is driven by an a priori assumption that European teams are more competitive. Future studies could explore label-free (unsupervised) methods to derive competitive balance.
- Further research might incorporate additional variables (such as financial data or injury rates) to enhance the understanding of competitiveness.